American Computer Science League

2023 Finals ● Short Problems ● Senior Division

1. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{\overline{A}\overline{B}}(AB \oplus \overline{B}C)$$

A. $AB + \overline{A}\,\overline{B}\,C$
B. $AB + A\,\overline{B}\,C$
C. $AB + \overline{A}\,B\,C$
D. $AB + \overline{A}\,\overline{B}\,\overline{C}$
E. None of the above

2. Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$\overline{A}(B \oplus \overline{C}) + B(\overline{A} + \overline{C})$$

A. 2
B. 3
C. 4
D. 5
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

(LCIRC-2 1101101) OR (RSHIFT-1 1011101) XOR 1101101 AND 1110111

A. 1011000
B. 1011011
C. 1110111
D. 1111111
E. None of the above

4. Bit-String Flicking

How many bit-string values of X (a bit-string of 5 bits) make the following equation TRUE?

(RCIRC-1 (LSHIFT-1 X)) OR NOT 10101 AND (LCIRC-1 (RSHIFT-2 X)) = 01011

A. 2
B. 3
C. 4
D. 5
E. None of the above

5. Recursive Functions

Function f is defined as follows for positive integers. Find the smallest value of x such that $f(x) > 256$.

$$f(x) = \begin{cases} 0 & \text{if } x = 1 \\ 1 & \text{if } x = 2 \\ 2 \cdot f(x-1) + f(x-2) & \text{if } x > 2 \end{cases}$$

A. 8
B. 9
C. 10
D. 11
E. None of the above

6. Recursive Functions

Find $f(f(54))$ if given:

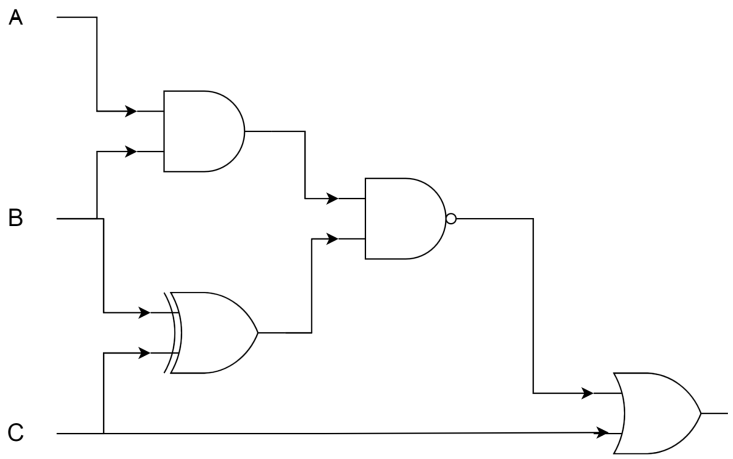
$$f(x) = \begin{cases} f\left(\left[\frac{x-4}{2}\right]\right) - 1 & \text{if } x \geq 18 \\ f(2 \cdot x + 1) + 3 & \text{if } 10 \leq x < 18 \\ x^2 & \text{if } x < 10 \end{cases}$$

Note: [a] is greatest integer less than or equal to a

A. 61
B. 83
C. 84
D. 89
E. None of the above

7. Digital Electronics

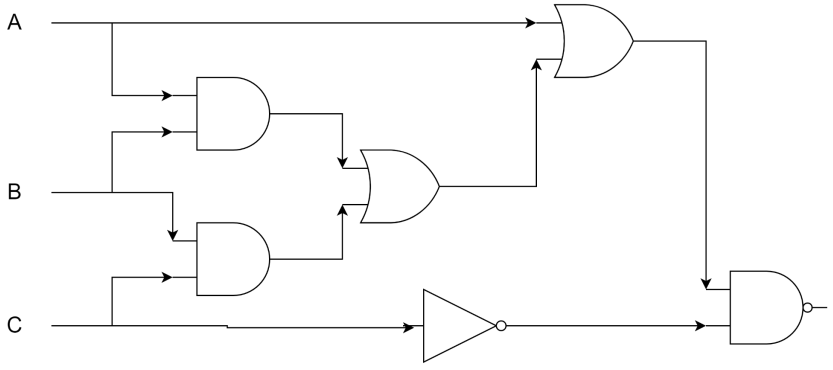
Which Boolean expression is equivalent to the following digital circuit?



A. $A + B + C$
B. $\overline{A} + B + C$
C. $A + \overline{B} + C$
D. $\overline{A} + \overline{B} + C$
E. None of the above

8. Digital Electronics

How many ordered triples make this digital circuit FALSE?



A. 0
B. 2
C. 4
D. 6
E. None of the above

9. Prefix-Infix-Postfix

Define a \$ b = $\left[\sqrt{|a * b|}\right]$ where [x] = the largest integer ≤ x. Evaluate the following postfix expression if all numbers are single digits:

6 4 \$ 2 5 \$ * 2 2 ^ − 2 4 ^ \$

A. 4
B. 11
C. 12
D. 16
E. None of the above

10. Prefix-Infix-Postfix

Given the equation of an ellipse with center (h,k) written in infix, convert it to prefix:

$(x - h)^2 / a^2 + (y - k)^2 / b^2 = 1$

A. /^−xh2^a2+/^−yk2^b2=1
B. =+^−xh2/^a2^−yk2/^b21
C. =+/^a2^−xh2/^b2^−yk21
D. =+/^−xh2^a2/^−yk2^b21
E. None of the above

11. Computer Number Systems

What is the 12th term of the Fibonacci sequence (1, 1, 2, 3, …) when converted to octal?

A. 144
B. 198
C. 220
D. 351
E. None of the above

12. Computer Number Systems

In the binary representation for the decimal numbers from 100 to 111, inclusive, how many times does the substring 101 occur in individual numbers?

A. 7
B. 8
C. 10
D. 12
E. None of the above

13. Data Structures

Build a min-heap for: **GREATHORNEDOWL**

List the nodes (left to right) at a depth of 2.

A. EEOL
B. NEGO
C. NEGL
D. NEHL
E. None of the above

14. Data Structures

To evaluate a postfix expression, apply the following steps for each character in the expression, from left to right:

if it is a number,
    push it onto the stack
if it is an operator, do the following stack operations:
    X=POP(), Y=POP()
    Z = perform the operator with (Y operator X)
    PUSH(Z)

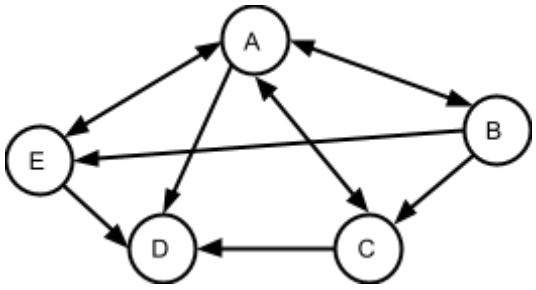
Given the following postfix expression, which of the choices represents every value that is placed onto the stack in the correct order?

3 2 ^ 5 − 4 8 + * 6 /

A. 3 2 9 5 4 4 8 12 48 6 8
B. 3 2 5 4 0 4 8 12 48 6 8
C. 3 2 8 5 -3 8 4 12 -36 6 -6
D. 3 2 9 5 4 4 8 12 48 6
E. None of the above

15. Graph Theory

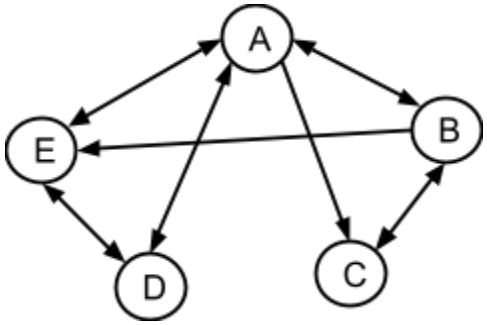
Given the directed graph below, what is the total number of paths of length 4?



A. 69
B. 76
C. 85
D. 91
E. None of the above

16. Graph Theory

How many cycles are there in the directed graph below that are of length 3 or more?



A. 4
B. 5
C. 7
D. 8
E. None of the above

17. What Does This Program Do?

Given that array **arr** contains the 10 values below, what is the final value of **count**?

8	6	2	9	3	5	7	1	4	0

```
count = 0
for x = 0 to 8
     for y = x+1 to 9
          if x == arr(y) then
               z1 = x
               z2 = y
               while z1 < z2
                    t = arr(z1)
                    arr(z1) = arr(z2)
                    arr(z2) = t
                    count = count + 1
                    z1 = z1 + 1
                    z2 = z2 - 1
               end while
          end if
     next y
next x
```

A. 14
B. 15
C. 16
D. 17
E. None of the above

18. LISP

What is the value of W when executing the following LISP program?

```
(SETQ X  '(a b (a c) (a (b c))))
(SETQ Y  '((c b) (a b c)))
(SETQ Z (CONS X Y))
(SETQ U (REVERSE Z))
(SETQ V (CADDR U))
(SETQ W (CADDR V))
```

A. (a c)
B. c
C. (c)
D. ((a c))
E. None of the above

19. FSA and Regular Expressions

Most regular expression libraries support the following notations:
\s matches a single whitespace (a space, tab, or new line character)
\S matches a single non-whitespace (a character other than a space, tab, or new line)

It's that time of year again to begin to choose courses for next semester. The registrar decided to challenge the students.
The course name must satisfy the following regular expression:

[^e] [a-hlr-y]* [\s]* [a-n]* [^mr]* [\S]* [0-9]*

How many courses from the following list can a student choose to enroll in this semester?

a) english	b) trig	c) python 8	d) french 2
e) java#8	f) c++	g) biology	h) adv physics

A. 3
B. 5
C. 6
D. 7
E. None of the above

20. Assembly Language

What is printed when the following program is run?

```
NUM     DC      440
M       DC      0
TOP1    LOAD    NUM
        SUB     =2
        BL      STOP
        LOAD    =2
        STORE   FACT
TOP2    LOAD    NUM
        DIV     FACT
        STORE   QUOT
        MULT    FACT
        SUB     NUM
        BE      NEXT
        LOAD    FACT
        ADD     =1
        STORE   FACT
        BU      TOP2
NEXT    LOAD    FACT
        SUB     M
        BL      CONT
        LOAD    FACT
        STORE   M
CONT    LOAD    QUOT
        STORE   NUM
        BU      TOP1
STOP    PRINT   M
        END
```

A. 3
B. 5
C. 7
D. 11
E. None of the above